

Validation of 2022 CPS Turnout Estimates Against McDonald VEP Figures

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2026-06-02

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1 Overview

This document compares state-level voter turnout estimates from the 2022 CPS Voting and Registration Supplement — using the **standard Census survey weight** (`WEIGHT`) — against the authoritative Voter Eligible Population (VEP) turnout figures from Dr. Michael McDonald’s United States Elections Project.

Using the standard weight rather than the Hur-Achen reweighted weight allows us to observe the raw discrepancy between what the CPS reports and the true turnout figures. Because the CPS is known to overreport turnout (due to social desirability bias and differential non-response), we expect CPS estimates to be systematically higher than McDonald’s VEP figures in most states. The size of this overestimation varies by state and is precisely what the Hur-Achen reweighting procedure corrects for.

The turnout variable used is `hurachen_turnout`, which drops respondents who answered Don’t Know, Refused, or No Response (treating them as missing rather than as non-voters). This isolates the effect of the **weight** on the discrepancy, holding the coding of the turnout variable constant.

2 CPS 2022 Turnout Estimates (Standard Weight)

3 McDonald VEP Turnout Estimates

4 State-by-State Comparison

The table below shows, for each state:

- **McDonald VEP (%)** — authoritative VEP turnout from the Elections Project

- **CPS Standard Weight (%)** — CPS estimate using the standard Census weight
- **Difference (pp)** — CPS minus McDonald (positive = CPS overestimates)

Differences greater than 5 percentage points are highlighted in red.

Table 1: 2022 State Voter Turnout: CPS Standard Weight vs. McDonald VEP Estimates

State	McDonald VEP (%)	CPS Std. Weight (%)	Difference (pp)
Alabama	37.3	56.7	19.4
Alaska	50.2	64.8	14.6
Arizona	49.3	66.3	17.0
Arkansas	41.5	53.4	11.9
California	42.9	62.7	19.8
Colorado	58.3	78.4	20.1
Connecticut	48.6	59.6	11.0
Delaware	42.8	62.8	20.0
District of Columbia	40.6	70.2	29.6
Florida	48.1	64.1	16.0
Georgia	51.9	68.1	16.2
Hawaii	40.5	59.3	18.8
Idaho	42.6	54.5	11.9
Illinois	45.8	63.4	17.6
Indiana	37.2	48.4	11.2
Iowa	51.9	58.3	6.4
Kansas	47.6	64.9	17.3
Kentucky	44.8	60.2	15.4
Louisiana	42.0	59.6	17.6
Maine	60.8	70.4	9.6
Maryland	45.8	68.7	22.9
Massachusetts	48.8	67.1	18.3
Michigan	58.9	74.0	15.1
Minnesota	60.4	72.5	12.1
Mississippi	32.5	53.2	20.7
Missouri	49.6	59.2	9.6
Montana	53.2	68.1	14.9
Nebraska	48.4	58.2	9.8
Nevada	45.8	62.8	17.0
New Hampshire	56.3	67.3	11.0
New Jersey	41.0	62.1	21.1
New Mexico	46.3	62.9	16.6
New York	42.5	62.1	19.6
North Carolina	47.5	61.5	14.0
North Dakota	41.7	57.0	15.3
Ohio	47.1	55.2	8.1
Oklahoma	39.6	53.0	13.4
Oregon	62.3	77.2	14.9
Pennsylvania	54.4	69.2	14.8
Rhode Island	43.6	62.8	19.2
South Carolina	42.8	56.5	13.7
South Dakota	53.1	61.2	8.1
Tennessee	33.3	51.2	17.9
Texas	41.8	55.4	13.6
Utah	46.8	62.1	15.3

Table 1: 2022 State Voter Turnout: CPS Standard Weight vs. McDonald
VEP Estimates (*continued*)

State	McDonald VEP (%)	CPS Std. Weight (%)	Difference (pp)
Vermont	56.0	72.7	16.7
Virginia	47.6	62.8	15.2
Washington	55.3	68.6	13.3
West Virginia	35.4	48.4	13.0
Wisconsin	59.8	68.8	9.0
Wyoming	45.6	55.7	10.1

5 Summary Statistics

Table 2: Summary of CPS (standard weight) vs. McDonald Differences (percentage points)

Statistic	Value
Mean difference (pp)	15.21
Median difference (pp)	15.20
SD of difference (pp)	4.35
Max overestimate (pp)	29.60
Min difference (pp)	6.40
States overestimating	51.00
States within 5pp	0.00
States > 5pp off	51.00

A positive mean difference confirms the well-documented CPS overreporting tendency. The mean and median discrepancy reflect the degree of bias that the Hur-Achen reweighting procedure corrects for. **Min difference** is the state where CPS overestimates the least (or, if negative, underestimates the most); a positive min difference means no state actually falls below McDonald’s VEP figure.

6 Largest Discrepancies

States differing by more than 5 percentage points are listed below, sorted by absolute discrepancy. These are the states where non-response bias or overreporting is most pronounced, and where the Hur-Achen reweighting makes the largest practical difference.

Table 3: States with discrepancy greater than 5 percentage points

State	McDonald VEP (%)	CPS Std. Weight (%)	Difference (pp)
District of Columbia	40.6	70.2	29.6
Maryland	45.8	68.7	22.9
New Jersey	41.0	62.1	21.1
Mississippi	32.5	53.2	20.7
Colorado	58.3	78.4	20.1
Delaware	42.8	62.8	20.0
California	42.9	62.7	19.8
New York	42.5	62.1	19.6
Alabama	37.3	56.7	19.4
Rhode Island	43.6	62.8	19.2
Hawaii	40.5	59.3	18.8
Massachusetts	48.8	67.1	18.3
Tennessee	33.3	51.2	17.9
Illinois	45.8	63.4	17.6
Louisiana	42.0	59.6	17.6
Kansas	47.6	64.9	17.3
Arizona	49.3	66.3	17.0
Nevada	45.8	62.8	17.0
Vermont	56.0	72.7	16.7
New Mexico	46.3	62.9	16.6

Georgia	51.9	68.1	16.2
Florida	48.1	64.1	16.0
Kentucky	44.8	60.2	15.4
Utah	46.8	62.1	15.3
North Dakota	41.7	57.0	15.3
Virginia	47.6	62.8	15.2
Michigan	58.9	74.0	15.1
Oregon	62.3	77.2	14.9
Montana	53.2	68.1	14.9
Pennsylvania	54.4	69.2	14.8
Alaska	50.2	64.8	14.6
North Carolina	47.5	61.5	14.0
South Carolina	42.8	56.5	13.7
Texas	41.8	55.4	13.6
Oklahoma	39.6	53.0	13.4
Washington	55.3	68.6	13.3
West Virginia	35.4	48.4	13.0
Minnesota	60.4	72.5	12.1
Arkansas	41.5	53.4	11.9
Idaho	42.6	54.5	11.9
Indiana	37.2	48.4	11.2
Connecticut	48.6	59.6	11.0
New Hampshire	56.3	67.3	11.0
Wyoming	45.6	55.7	10.1
Nebraska	48.4	58.2	9.8
Maine	60.8	70.4	9.6
Missouri	49.6	59.2	9.6
Wisconsin	59.8	68.8	9.0
Ohio	47.1	55.2	8.1
South Dakota	53.1	61.2	8.1
Iowa	51.9	58.3	6.4

7 Notes

- The standard Census weight (**WEIGHT**) is used here, **not** the Hur-Achen reweighted weight (**turnout_weight**). The reweighted weight is specifically designed to eliminate these discrepancies by post-stratifying to McDonald’s VEP figures; using it here would produce near-perfect agreement by construction and would not illuminate the underlying bias.
- The **hurachen_turnout** coding is used (DK/Refused/No Response treated as missing). To see the additional effect of the Census’s own coding (which treats these as No), substitute **cps_turnout** for **hurachen_turnout**.
- McDonald’s VEP data for 2022 retrieved from: <https://docs.google.com/spreadsheets/d/17iSZIBPP6jj0C2wcqpym9>

CPS vs. McDonald Turnout Estimates, 2022

Using standard Census survey weight; points above the line = CPS overestimates

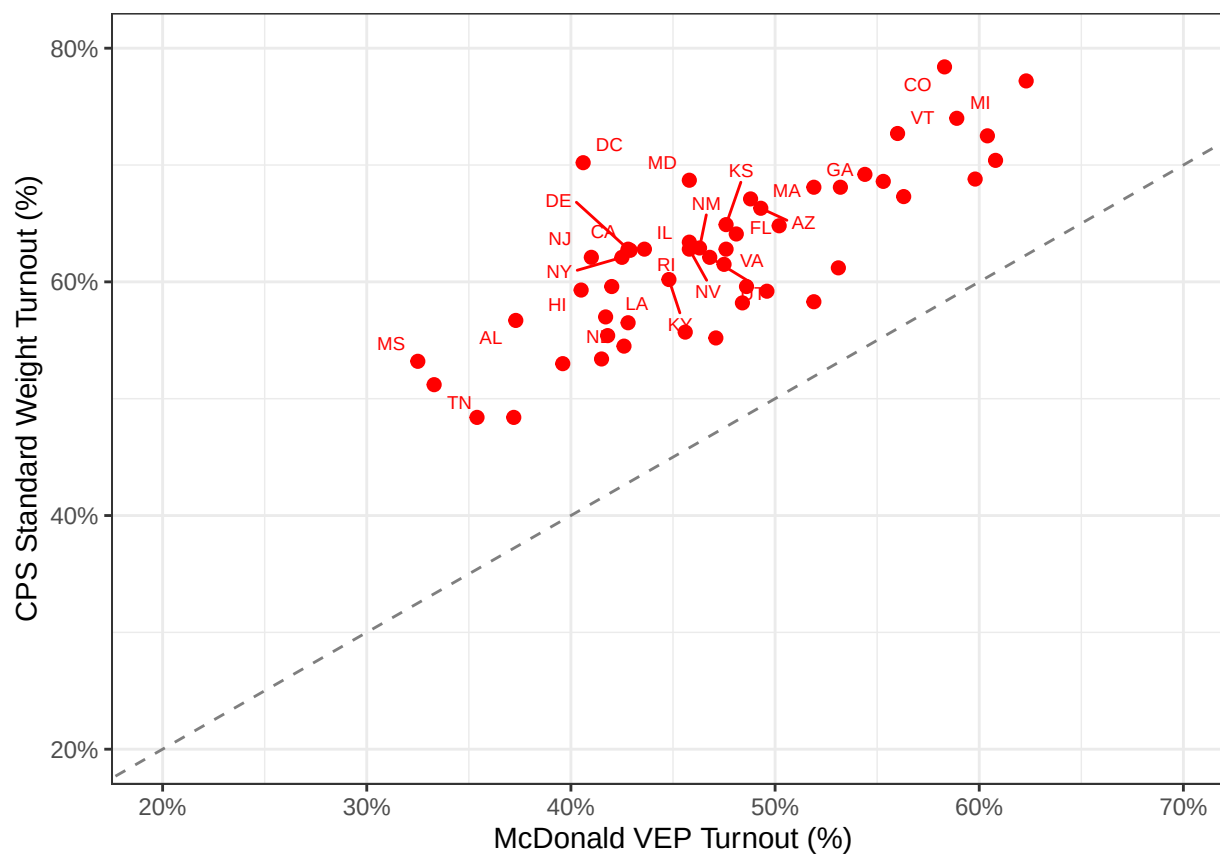


Figure 1: CPS (standard weight) vs. McDonald VEP turnout by state, 2022. The dashed line is the 45-degree line of perfect agreement. Points above the line indicate CPS overestimation. States labeled in red differ by more than 5 percentage points.